

Sam Bolger

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EDUCATION

Concordia College

Moorhead, MN

Bachelor of Science in Data Science, Minors in Mathematics and Cybersecurity

2022 – 2026

- Activities and Awards: Dean's List, Data Science Club, Concordia Men's Basketball

EXPERIENCE

Enterprise Solution Architecture Analyst Intern (AI Engineering)

May 2026 – Present

Entegris

Chaska, MN

- Built and deployed a full-stack RAG web app using Python, FastAPI, React, TypeScript, and Vertex AI on GCP.
- Cut architecture pattern drafting from about two weeks to under a minute using BM25 retrieval over 500+ docs.
- Designed an LLM eval harness measuring retrieval hit rate, faithfulness, and hallucination rate to catch regressions.
- Automated a manual application ID entry workflow with GitHub Actions, saving weekly hours for architects.
- Partnered with cross-functional teams to drive architectural decisions and meet business goals.

Data Science Associate

Jan. 2025 – May 2025

Concordia College Data Science Lab

Moorhead, MN

- Trained a k-means machine learning model on 12,000+ cafe transactions to surface customer buying patterns.
- Shipped a demand forecasting model with a Streamlit dashboard that informed staffing and inventory decisions.
- Presented findings and recommendations to non-technical stakeholders including cafe management and finance staff.

Parks Maintenance

June 2022 – Aug. 2025

City of Apple Valley

Apple Valley, MN

- Maintained city parks and athletic fields while training seasonal employees on equipment usage and safety protocols.
- Consistently completed daily routes ahead of schedule and mentored new hires over four consecutive summers.

PROJECTS

PR Reviewer Agent | *Python, FastAPI, LangGraph, Anthropic Claude, Chroma, MCP, Fly.io*

[GitHub](#)

- Deployed an agentic GitHub App that reviews PRs using LangGraph, RAG, and structured Claude tool use.
- Designed a signal-derived confidence score using citation validity, RAG hits, and model self-report.
- Wrote an LLM-as-judge eval harness with prompt A/B testing, MCP server, and 66 pytest cases in CI/CD.

Tech Brief | *Python, dbt, DuckDB, LangGraph, Anthropic Claude, GitHub Actions, Resend*

[GitHub](#)

- Engineered a daily ELT pipeline ingesting HackerNews, Reddit, RSS, GitHub trending, and ArXiv into DuckDB.
- Structured dbt medallion models with schema tests, scoring stories by source weight, recency, and engagement.
- Wrote a LangGraph agent using Claude Haiku for structured summaries, delivered daily via Resend.

Bat Echolocation Detection | *Python, TensorFlow, MobileNetV2, librosa*

[GitHub](#)

- Preprocessed 33,000+ ultrasonic audio files into spectrograms with librosa for supervised classification.
- Trained a MobileNetV2 CNN in TensorFlow with transfer learning, achieving 96.4% recall on bat calls.
- Deployed an interactive web app for real-time bat-call inference with per-class confidence scores.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, R, HTML/CSS

Libraries: FastAPI, dbt, React, Chroma, Streamlit, Pandas, NumPy

Machine Learning: Deep Learning, Transfer Learning, TensorFlow, scikit-learn, XGBoost, MLflow

Cloud & Tools: GCP, Vertex AI, Azure, AWS, Docker, GitHub Actions, CI/CD, ETL/ELT, DuckDB, PostgreSQL

AI & LLMs: LangGraph, LangChain, RAG, Agentic AI, MCP, Prompt Engineering, Claude, OpenAI, Vector Databases